DELIVERABLE 2 - TODO 2

In Deliverable 1 (D1), the expert system assumes that all sensor readings are fully reliable.
For example, if (co-alarm on) is observed, the system immediately triggers a full emergency.

However, in real-world smart home environments, sensor readings are inherently uncertain:

- CO detectors may produce false positives
- Smoke sensors may be triggered by cooking
- Occupancy sensors may misdetect presence
- Air quality sensors may vary due to calibration or location

To address this, Deliverable 2 introduces Certainty Factors (CF) to model probabilistic uncertainty.

A certainty factor is defined as: CF ∈ [0.0, 1.0], CF = 1.0 → fully certain and CF = 0.0 → no confidence. This allows the system to reason not only about what happened, but also how reliable that observation is.

To support CF reasoning, the env template is extended with five new slots:

```
(slot co-alarm-cf (type FLOAT) (range 0.0 1.0))
(slot fire-alarm-cf (type FLOAT) (range 0.0 1.0))
(slot occupancy-cf (type FLOAT) (range 0.0 1.0))
(slot iaqi-cf (type FLOAT) (range 0.0 1.0))
(slot aqhi-cf (type FLOAT) (range 0.0 1.0))
```

Each slot represents the confidence level of a corresponding sensor reading. Example:

```
(env (date "2026-02-15")
    (co-alarm on)
    (co-alarm-cf 0.85))
```
← the CO alarm is triggered with 85% confidence.

(Across all days in facts.clp, these additional CF fields significantly expand the factbase, satisfying the requirement of ≥5 new useful facts.)

CF-Based Reasoning Strategy:

Instead of binary decisions, the system now uses threshold-based reasoning:

CF Range	Interpretation	Action
CF ≥ 0.70	High confidence	Full action
0.30 ≤ CF < 0.70	Moderate confidence	Cautious response
CF < 0.30	Low confidence	Likely false alarm

Implementation in Rulebase:

10 new CF rules have been added.

4.1 High Confidence Emergency

```
(defrule co-high-cf
   (env (co-alarm on) (co-alarm-cf ?cf))
   (test (>= ?cf 0.70))
   =>
   (assert (emergency))
)
```

- Replaces D1 behavior
- Only triggers emergency when confidence is high

4.2 Moderate Confidence Warning

```
(defrule co-moderate-cf
   (env (co-alarm on) (co-alarm-cf ?cf))
   (test (and (>= ?cf 0.30) (< ?cf 0.70)))
   =>
   (assert (msg "... investigate ..."))
)
```

- Avoids unnecessary panic
- Introducing graduate response

4.3 Low Confidence Handling

```
(defrule co-low-cf
   (env (co-alarm on) (co-alarm-cf ?cf))
   (test (< ?cf 0.30))
   =>
   (assert (msg "... likely false positive ..."))
)
```

- Improves system reliability
- Prevents unnecessary interventions

4.4 Combining Evidence

```
(bind ?combined (+ ?cf1 (* ?cf2 (- 1 ?cf1))))
```

This implements:

CF_combined = CF1 + CF2 * (1 - CF1)

Example:

- CO CF = 0.85
- Fire CF = 0.60 → Combined CF = 0.94

This enables multi-sensor reasoning, improving decision accuracy.

4.5 Handling Uncertain Sensors

Additional rules include:

- occupancy-uncertain-thermostat → fallback to energy-saving mode
- iaqi-uncertain-reading → qualify air quality recommendation
- aqhi-uncertain-reading → downgrade trust in outdoor data

DELIVERABLE 2 - TODO 3

While Certainty Factors model uncertainty in reliability, they do not address vagueness in concepts.

Example: Is 19°C “cold” or “comfortable”?

In D1, the system uses hard thresholds:

temp < 20 → heater ON

This creates abrupt behavior.

To address this, D2 introduces Fuzzy Logic, which models concepts as gradual transitions.

Extension of Factbase:

New template:

```
(deftemplate fuzzy-env

   (slot mu-temp-cold)

   (slot mu-temp-cool)

   (slot mu-temp-comfortable)

   (slot mu-temp-warm)
```

```
    (slot mu-temp-hot)

    ...

)
```

Each slot represents a membership degree: $\mu \in [0.0, 1.0]$

Example:

```
(mu-temp-cold 0.3)
(mu-temp-comfortable 0.7)
```

This means: The temperature is partially cold and mostly comfortable.

Each day now includes:

```
(fuzzy-env (date "..."))
```

Thus, satisfying the requirement of ≥ 5 new facts.

Fuzzification:

Crisp values are converted into fuzzy memberships using: (deffunction mu-trapezoid ...)

Example rule:

```
(defrule fuzzify-temperature
    (env (temp ?t))
    =>
    (bind ?cold (mu-trapezoid ?t -999 -999 14 18))
)
```

This transforms:

temp = 19°C

into:

$\mu(\text{cold}) \approx 0.25$
$\mu(\text{comfortable}) \approx 0.75$

Fuzzy Inference:

We use linguistic reasoning:

```
(defrule fuzzy-heat-strong
```

```
    (fuzzy-env (mu-temp-cold ?mu))

    (test (> ?mu 0.0))

    =>

    (assert (msg "... strong heating ..."))

)
```

Interpretation: If temperature is cold to some degree, heating should be applied proportionally.

Defuzzification:

To convert fuzzy outputs into a concrete action: (bind ?target (/ ?numerator ?denominator))

This implements the **centroid method**: target = Σ(μ × value) / Σ(μ)

Example:

- cold → 21°C (μ=0.7)
- cool → 19°C (μ=0.4)

→ result ≈ 20°C

Then:

(modify thermostat (target-temp 20))

Additional Fuzzy Rules:

- fuzzy-humidifier-strong
- fuzzy-humidifier-mild
- fuzzy-dehumidifier
- fuzzy-window-poor-aqhi
- fuzzy-window-moderate-aqhi

These provide:

- Gradual humidity control
- Soft air-quality recommendations

DELIVERABLE 2 - TODO 4

1. Selected Quality Attributes

1. Explainability

2. Reliability

3. Accuracy

4. Maintainability

These attributes are chosen because Deliverable 2 introduces uncertainty (both probabilistic and possibilistic), which directly affects how decisions are made, justified, and trusted.

1.1 Explainability

The system makes decisions that affect physical devices (e.g., thermostat, ventilation, alarms).

Therefore, users must be able to understand:

- why a decision was made
- what data triggered it
- how confident the system is

Without explainability, the system would not be trustworthy.

1.2 Reliability

In D1, all sensor readings were assumed correct.

In reality:

- CO alarms can fail
- occupancy sensors can misdetect
- air quality readings may be inaccurate

Thus, reliability becomes critical when introducing uncertain knowledge (CF).

1.3 Accuracy

D1 used hard thresholds (e.g., temp < 20).

This causes unrealistic behavior:

- 19.9°C → heater ON
- 20.0°C → heater OFF

Fuzzy Logic in D2 aims to improve accuracy of modeling real-world conditions, especially human comfort.

1.4 Maintainability

Smart home systems evolve:

- new sensors may be added
- thresholds may change
- user preferences may differ

Thus, the knowledge base must be easy to modify without rewriting the entire system.

2. Application of Quality Guidelines

2.1 Explainability

Where applied:

- Every rule (D1 + CF + Fuzzy) asserts a message: (assert (msg (date ?date) (text "...")))
- CF rules include confidence values: "EMERGENCY (CF=0.85): CO detected..."
- Fuzzy rules include membership values: "membership in 'cold' = 0.7"
- Defuzzification explains calculation: "thermostat set to 20C (cold=0.6 cool=0.3 ...)"
- Output is grouped by date using: (print-grouped)

Guideline applied

- “Support explainability through traceable reasoning”
- “Every action must have justification”

2.2 Reliability

Where applied:

- CF-based emergency handling:
- co-high-cf
- co-moderate-cf
- co-low-cf
- fire-*

Combined evidence rule: co-and-fire-combined-cf

Sensor uncertainty handling:

- occupancy-uncertain-thermostat
- iaqi-uncertain-reading
- aqhi-uncertain-reading

Guideline applied

- “Do not assume facts are always correct”
- “Handle uncertainty explicitly”

2.3 Accuracy

Where applied:

Fuzzification rules:

- fuzzify-temperature
- fuzzify-humidity
- fuzzify-aqhi

Membership functions: (deffunction mu-trapezoid ...)

Fuzzy inference:

- fuzzy-heat-strong
- fuzzy-heat-moderate

Defuzzification: fuzzy-defuzzify-thermostat

Guideline applied

- “Avoid sharp thresholds when modeling real-world concepts”
- “Use gradual transitions where appropriate”

2.4 Maintainability

Where applied:

Separation of files:

- templates.clp
- facts.clp
- rules.clp
- run.clp

Modular templates:

- env → raw data
- fuzzy-env → derived fuzzy data

Reusable function: (deffunction mu-trapezoid ...)

Thresholds clearly visible in rules: (test (>= ?cf 0.70))

Guideline applied

- “Separate facts and rules”
- “Promote modularity”
- “Avoid magic numbers scattered across code”

3. Impact Analysis:

3.1 Explainability

Positive impact

- Users can trace every decision
- CF values provide transparency
- Fuzzy membership gives deeper insight

Example:

Instead of: CO detected → EMERGENCY

Now: CO detected (CF=0.45) → WARNING

This significantly improves trust.

3.2 Reliability

Positive impact

- System avoids false alarms
- Handles sensor noise realistically
- Supports safer decision-making

Example:

- D1: always emergency
- D2: depends on confidence

3.3 Accuracy

Positive impact

- Smooth control instead of binary jumps
- More realistic temperature regulation
- Better alignment with human comfort

Example:

- D1: fixed 20°C

- D2: computed 18–21°C depending on conditions

3.4 Maintainability

Mixed (positive with minor trade-off)

Positive

- Easy to tune thresholds
- Easy to add new fuzzy variables
- Code structure is clean

Negative (minor)

- More parameters (CF values, fuzzy boundaries)
- Slightly harder to read than D1

Overall: strong positive impact

Summary:

The integration of Certainty Factors and Fuzzy Logic not only improves decision-making under uncertainty, but also enhances the overall quality of the expert system in terms of explainability, reliability, accuracy, and maintainability.